

CHELLAMUTHU S

chellamuthu5824@gmail.com | [linkedin.com/in/chellamuthu-s](https://www.linkedin.com/in/chellamuthu-s) | github.com/chellamuthu | chellamuthu.github.io/portfolio | Pudukkottai, India | +91 9488645824

OBJECTIVE

Final-year B.E. Computer Science and Engineering student with hands-on full-stack development experience (Node.js, Express, SQL Server, REST APIs) and a strong record of building AI-driven and embedded systems, including a real-time EV telemetry platform (ESP32, CAN Bus, LoRa) and an AI-powered coding assessment platform with local LLM-based question validation (Ollama, Judge0). Working knowledge of Python, Django, and Retrieval-Augmented Generation (RAG) pipelines using Sentence Transformers and Groq/LLM APIs. Seeking an internship to apply strong analytical and problem-solving skills to production-grade enterprise AI and cloud-native engineering challenges in a fast-paced startup environment.

EDUCATION

Bachelor of Technology — Computer Science & Engineering

2023 – 2027 (Pursuing)

Mount Zion College of Engineering and Technology, Pudukkottai | CGPA: 7.04 / 10

Class XII — State Board | 2023 | 70% — Govt Hr Sec School, Arimalam

TECHNICAL SKILLS

Languages: Java (Core + OOP), Python, JavaScript, SQL, HTML/CSS

Frameworks: Node.js, Express.js, Django, REST APIs, MVC Pattern

Embedded / IoT: ESP32, CAN Bus, LoRa, sensor integration, real-time telemetry systems

AI / GenAI: Ollama (local LLM inference), RAG pipelines, Sentence Transformers, Groq API, prompt engineering, Judge0 integration, AI-based logic verification

Mobile: Flutter (real-time data visualization apps)

Databases: Microsoft SQL Server (MSSQL), MySQL, PostgreSQL, SQLite

Tools & DevOps: Git, GitHub, JWT-based SSO, Postman, VS Code, IntelliJ IDEA

PROJECTS

GoKart Live Telemetry Dashboard — Zion Racing EV GoKart Team

2025 – Present

ESP32, CAN Bus, LoRa, Node.js, Flutter

- Serving as Telemetry Head and Innovation Lead, built a live telemetry system streaming CAN Bus vehicle data over LoRa to a Node.js backend and Flutter dashboard.
- Contributed to the team winning Runner-Up for Best EV Innovation at FKRC Season 6.

MZAptiCode+ — College Coding & Aptitude Assessment Platform

2025 – Present

Node.js, Express, SQL Server, Ollama, Judge0

- Built a full-stack exam and coding-practice platform with topic/level progression, Judge0-based code execution, and JWT-based Single Sign-On across multiple college portal apps.
- Designed an AI-powered question generation and self-correction pipeline using Ollama (qwen3:8b), with a two-stage submission gate that runs test cases and an AI logic check to detect hardcoding or bypassed solutions.
- Resolved server scalability issues under concurrent load and built offline-resilient exam submission using a localStorage queue with automatic flush on reconnect.

Tamil Sign Language Recognition System

2025

Computer Vision, Machine Learning

- Built a real-time Tamil sign language recognition system; won the Novel Solution Award at Make-A-Thon '25.

HR Employee Chatbot (EmpQuery)

2024 – 2025

Django, RAG, Sentence Transformers, Groq API

- Built a Django-based Retrieval-Augmented Generation chatbot using Sentence Transformers embeddings with cosine similarity search, later optimized for speed with the Groq API (llama3-8b-8192).
- Implemented a two-call architecture separating intent classification from answer generation, with session-based context for follow-up questions.

EXPERIENCE

Full Stack & AI Development Intern — White House Business Solutions

2025

- Built a Timetable Management System and a Hospital Management System using Django and PostgreSQL.
- Developed the EmpQuery RAG-based HR chatbot, including intent classification and grounded answer generation.

ACHIEVEMENTS & ACTIVITIES

- Telemetry Head & Innovation Lead, Zion Racing EV GoKart Team.
- Winner, Novel Solution Award — Make-A-Thon '25 (Tamil Sign Language Recognition).
- Runner-Up, Best EV Innovation — FKRC Season 6 (Zion Racing EV GoKart Team).
- GitHub: github.com/chellamuthu